

14.2

18. **Three Entrances** A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance I.

Entrance	1	2	3
Number Entering	83	61	56

$n = 200$

$$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$$

H_a : At least one entrance proportion is different

$$\alpha = .05, \quad \alpha_1 = .01, \quad \alpha_2 = .04, \quad n = 200$$

$$E_i = np_i$$

$$E_1 = 200 \left(\frac{1}{3} \right) = 66.67$$

$$E_2 = 200 \left(\frac{1}{3} \right) = 66.67$$

$$E_3 = 200 \left(\frac{1}{3} \right) = 66.67$$

$$\chi^2_{\text{test}} = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67} = 5.99 + 0.48 + 1.61 = 8.08$$

$$df: k-1 = 3-1 = 2$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, k-1} = 5.991$$

$$R.R.: \chi^2 > 5.991$$

$$8.08 > 5.991 \text{ (Reject } H_0)$$

→ There is significant evidence at 0.05 significance level

to conclude that the three entrances are not equally preferred

95% CI for entrance 1 proportion

$$\hat{p} = \frac{83}{200} = 0.415$$

$$CI: \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.415 \pm 1.96 \sqrt{\frac{0.415(0.585)}{200}}$$

$$0.415 \pm 0.0605$$

$$(0.397, 0.483)$$

22. **Heart Attacks on Mondays** Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day. In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded.

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$$H_0: p_{\text{Sun}} = p_{\text{Mon}} = p_{\text{Tue}} = p_{\text{Wed}} = p_{\text{Thu}} = p_{\text{Fri}} = p_{\text{Sat}} = \frac{1}{7}$$

H_a : At least one day has a different proportion

$$E_i = n \left(\frac{1}{7} \right) = \frac{200}{7} = 28.571$$

$$\chi^2_{\text{test}} = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(24-28.571)^2}{28.571} + \frac{(36-28.571)^2}{28.571} + \frac{(27-28.571)^2}{28.571} + \frac{(26-28.571)^2}{28.571} + \frac{(32-28.571)^2}{28.571} + \frac{(26-28.571)^2}{28.571} + \frac{(29-28.571)^2}{28.571}$$

$$= 0.791 + 1.932 + 0.086 + 0.291 + 0.412 + 0.291 + 0.006 = 3.414$$

$$df: k-1 = 7-1 = 6$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, k-1} = 12.592$$

$$R.R.: \chi^2 > 12.592$$

$$3.414 < 12.592 \text{ (Fail to reject } H_0)$$

→ There is no significant evidence at 0.05 significance level to conclude that the incidence of heart attacks differ by day of the week.

14.3

18. **Fatal Accidents** Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	Total
Fatal	67	26	16	99
Not Fatal	128	63	46	237

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0 : Vehicle and automobile size are independent

H_a : Vehicle and automobile size are dependent

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$$

$$E_{11} = \frac{99 \times 195}{346} = 56.95, \quad E_{12} = \frac{99 \times 151}{346} = 43.57$$

$$E_{21} = \frac{237 \times 195}{346} = 133.05, \quad E_{22} = \frac{237 \times 151}{346} = 103.46$$

$$E_{13} = \frac{99 \times 90}{346} = 25.55, \quad E_{23} = \frac{237 \times 90}{346} = 61.47$$

$$\chi^2_{\text{test}} = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(67-56.95)^2}{56.95} + \frac{(26-43.57)^2}{43.57} + \frac{(16-25.55)^2}{25.55} + \frac{(128-133.05)^2}{133.05} + \frac{(63-103.46)^2}{103.46} + \frac{(46-61.47)^2}{61.47}$$

$$= 1.505 + 0.197 + 0.150 + 0.192 + 0.068 + 0.195 = 1.803$$

$$df: (r-1)(c-1) = (2-1)(3-1) = 2$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, k-1} = 5.991$$

$$R.R.: \chi^2 > 5.991$$

$$1.803 < 5.991 \text{ (Fail to reject } H_0)$$

→ There is no significant evidence at 0.05 significance level to conclude that the frequency of fatal accidents depends on automobile size

23. **Telecommuting** As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 500 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week			Total
	Less Than One	At Least One, Not More Than 3	All at Home	
Under \$10,000	18	16	14	48
\$10,000-\$14,999	14	20	12	46
\$15,000-\$19,999	15	22	9	46
Above \$20,000	13	20	12	45

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusion from part a?

a) H_0 : Salary and number of workdays at home are independent

H_a : Salary and number of workdays at home are dependent

Expected table

	Less Than One	At Least One, Not More Than 3	All at Home
Under \$10,000	18.74	21.00	10.26
\$10,000-\$14,999	14.07	20.54	11.41
\$15,000-\$19,999	15.20	20.46	10.34
Above \$20,000	14.77	22.94	11.59

$$\chi^2_{\text{test}} = \sum \frac{(O_i - E_i)^2}{E_i} = 5.99$$

$$df: (r-1)(c-1) = (4-1)(3-1) = 6$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, k-1} = 12.592$$

$$R.R.: \chi^2 > 12.592$$

$$5.99 < 12.592 \text{ (Fail to reject } H_0)$$

→ There is no significant evidence at 0.05 significance level to conclude that salary is dependent on number of workdays spent at home

$$b) df = 6$$

$$\chi^2 = 5.99$$

$$p \approx 0.40 \text{ since } p > 0.05$$

→ Yes, the p-value confirms the conclusion that we do not reject H_0 .

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960! Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married		Total
	Yes	No	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	160	140	300
Total	509	491	1000

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H₀: Marriage rates are the same for all educational levels
H_a: Marriage rates differ among educational levels

$$E_{10} = \frac{300 \times 509}{1000} = 152.7$$

$$E_{11} = \frac{300 \times 491}{1000} = 147.3$$

$$\text{Total: } X^2 = \sum \frac{(O-E)^2}{E}$$

$$X^2 = \frac{(187-152.7)^2}{152.7} + \frac{(113-117.3)^2}{117.3} + \frac{(162-147.3)^2}{147.3} + \frac{(138-152.7)^2}{152.7} + \frac{(160-147.3)^2}{147.3} + \frac{(140-152.7)^2}{152.7}$$

$$= 7.597 + 0.346 + 1.791 + 3.291 + 0.114 + 1.177$$

$$= 10.06$$

$$df = (R-1)(C-1) = (3-1)(2-1) = 2$$

$$\alpha = 0.01, df = 2 \rightarrow X^2_{0.01,2} = 9.210$$

$$R.R. = X^2 > 9.210 \text{ (Reject H}_0\text{)}$$

→ There is sufficient evidence at 0.1 significance level to conclude that marriage rates differ significantly among educational levels.

→ People with higher education levels appear more likely to be married.

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence			Total
	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	0	24
3	10	10	14	34
4 or more	6	2	16	24
Total	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H₀: The family size distributions are the same for all residence types

H_a: At least one residence type has a different family size distribution

Expected Table

Members	Apartment	Duplex	Single Residence	Total
1	9.667	9.667	9.667	29.000
2	11	11	11	33.000
3	11.333	11.333	11.333	34.000
4+	8	8	8	24.000
Total	40	40	40	120

$$\text{Total: } X^2 = \sum \frac{(O-E)^2}{E} = 36.44$$

$$df = (R-1)(C-1) = (4-1)(3-1) = 6$$

$$\alpha = 0.01, df = 6 \rightarrow X^2_{0.01,6} = 16.812$$

$$R.R. = X^2 > 16.812$$

$$36.44 > 16.812 \text{ (Reject H}_0\text{)}$$

→ There is sufficient evidence at 0.01 significance level to conclude that family size distributions differ among the three residence types.

Interpretation

- Duplexes have many more 1-person household than expected
 - Single residences have many more large families (4+more) than expected
 - Apartment are more concentrated around 1-3 family members
- In residence types appears related to size of household.